Experiment 2: Data Visualization using Matplotlib and Seaborn

Implementation

Dataset Used: Titanic-Dataset.csv

Part A: Import Required Libraries

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Part B: Load Titanic Dataset

```
df = pd.read_csv("/content/Titanic-Dataset.csv")

print("Dataset Shape:", df.shape)
print(df.head())
```

```
Dataset Shape: (891, 12)
   PassengerId  Survived  Pclass  \
0            1         0       3
1            2         1       1
2            3         1       3
3            4         1       1
4            5         0       3

                                                Name     Sex   Age  SibSp  \
0                            Braund, Mr. Owen Harris    male  22.0      1
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0      1
2                             Heikkinen, Miss. Laina  female  26.0      0
3       Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0      1
4                           Allen, Mr. William Henry    male  35.0      0

   Parch            Ticket     Fare Cabin Embarked
0      0         A/5 21171   7.2500   NaN        S
1      0          PC 17599  71.2833   C85        C
2      0  STON/O2. 3101282   7.9250   NaN        S
3      0            113803  53.1000  C123        S
4      0            373450   8.0500   NaN        S
```

Part C: Basic Dataset Information

```
print(df.info())
print("Summary Statistics:\n", df.describe())
```

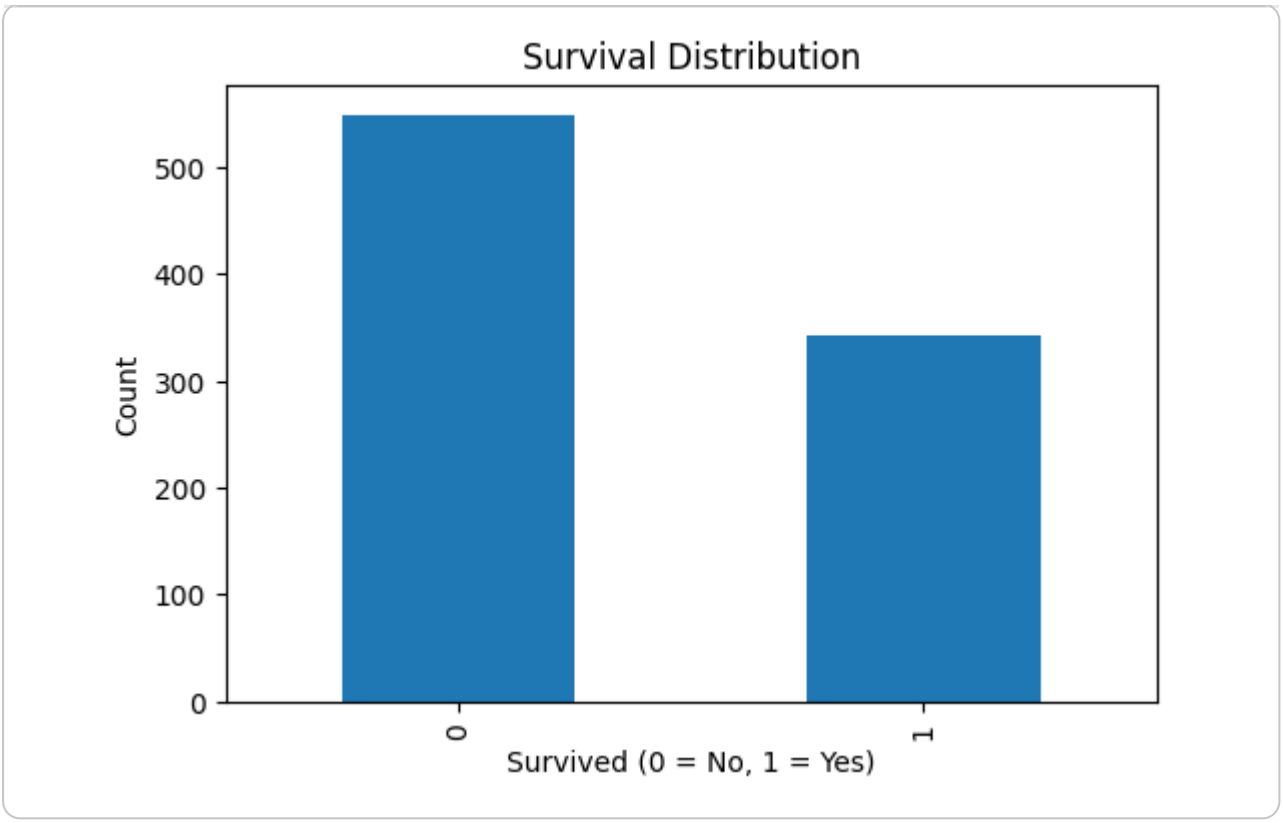
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
```

```
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
None
Summary Statistics:
        PassengerId    Survived      Pclass         Age       SibSp  \
count   891.000000  891.000000  891.000000  714.000000  891.000000
mean    446.000000    0.383838    2.308642   29.699118    0.523008
std     257.353842    0.486592    0.836071   14.526497    1.102743
min       1.000000    0.000000    1.000000    0.420000    0.000000
25%     223.500000    0.000000    2.000000   20.125000    0.000000
50%     446.000000    0.000000    3.000000   28.000000    0.000000
75%     668.500000    1.000000    3.000000   38.000000    1.000000
max     891.000000    1.000000    3.000000   80.000000    8.000000

            Parch        Fare
count  891.000000  891.000000
mean     0.381594   32.204208
std      0.806057   49.693429
min      0.000000    0.000000
25%      0.000000    7.910400
50%      0.000000   14.454200
75%      0.000000   31.000000
max      6.000000  512.329200
```

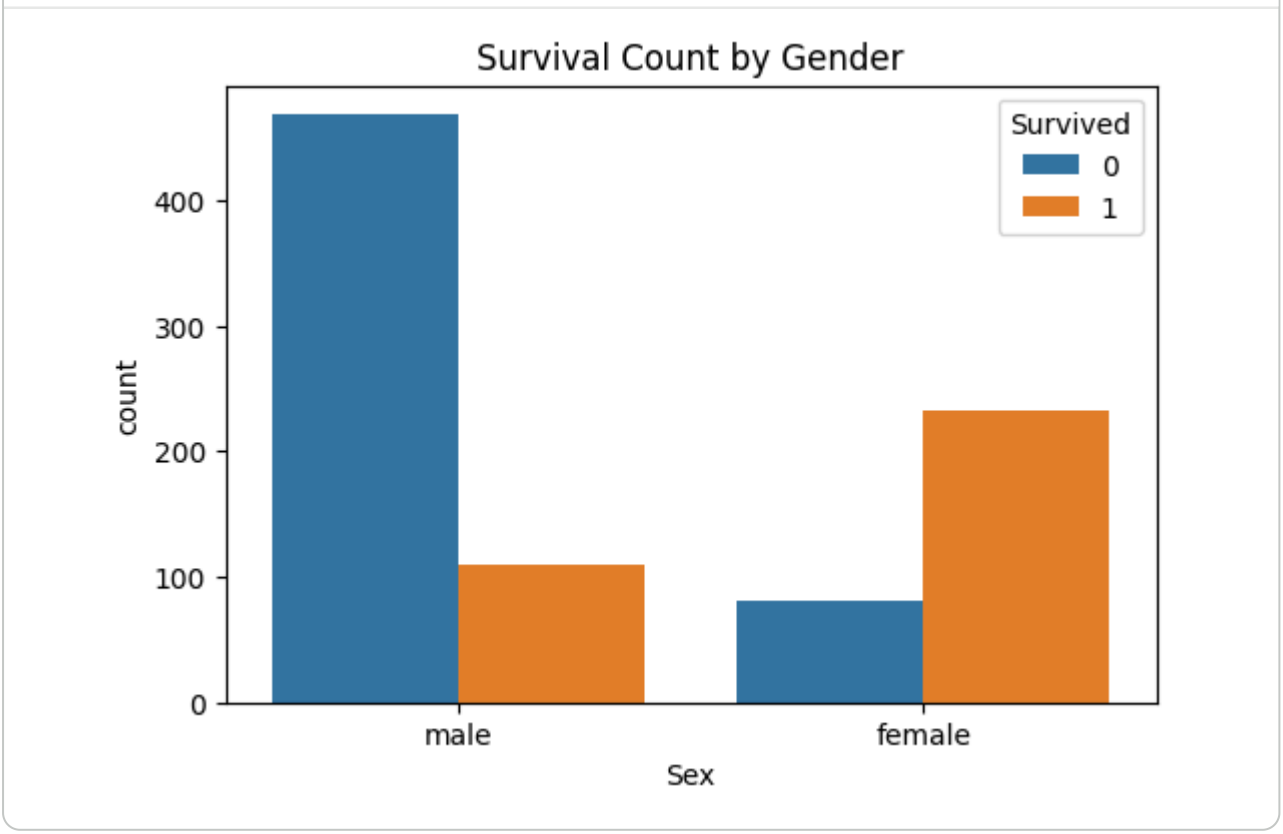
Part D: Bar Chart – Survival Distribution

```
plt.figure(figsize=(6,4))
df["Survived"].value_counts().plot(kind="bar")
plt.title("Survival Distribution")
plt.xlabel("Survived (0 = No, 1 = Yes)")
plt.ylabel("Count")
plt.show()
```



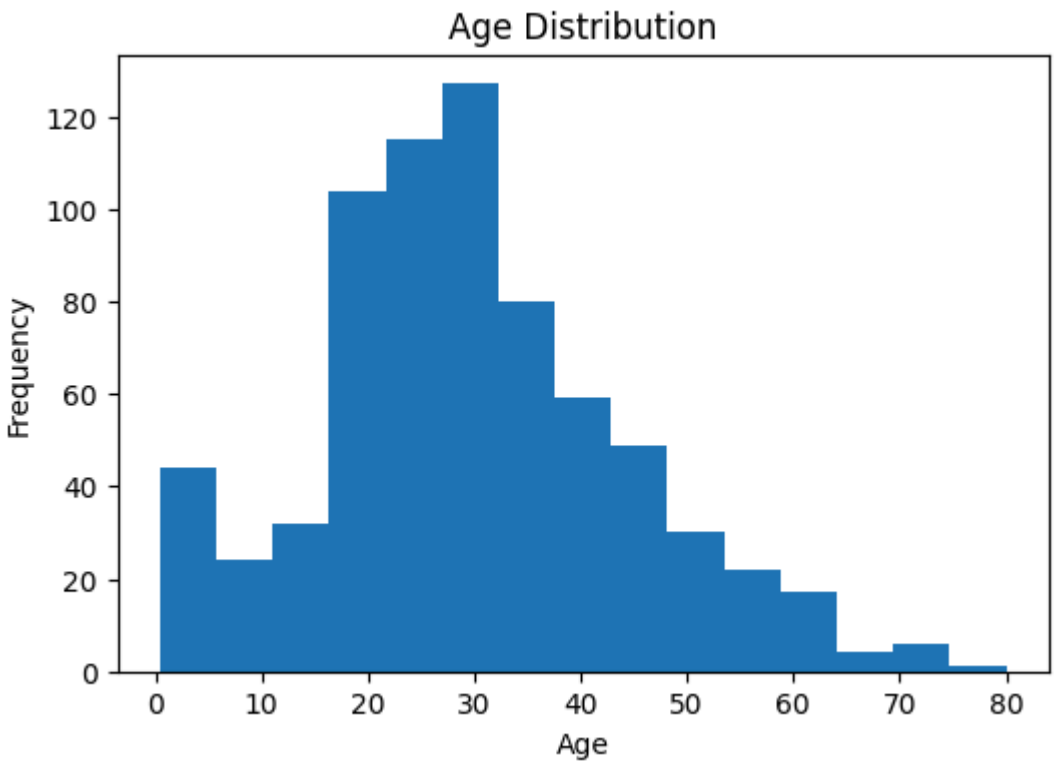
Part E: Count Plot – Survival by Gender

```
plt.figure(figsize=(6,4))
sns.countplot(x="Sex", hue="Survived", data=df)
plt.title("Survival Count by Gender")
plt.show()
```



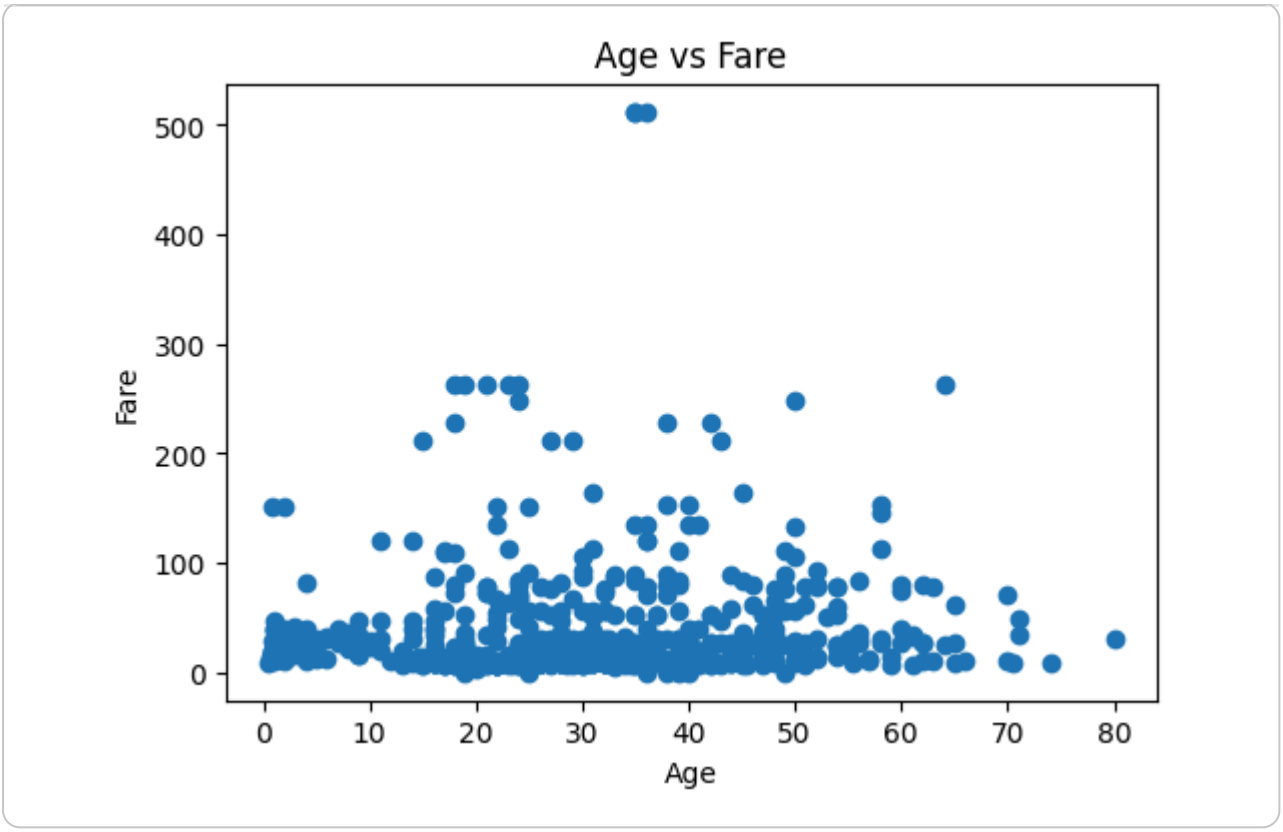
Part F: Histogram – Age Distribution

```
plt.figure(figsize=(6,4))
plt.hist(df["Age"].dropna(), bins=15)
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.show()
```



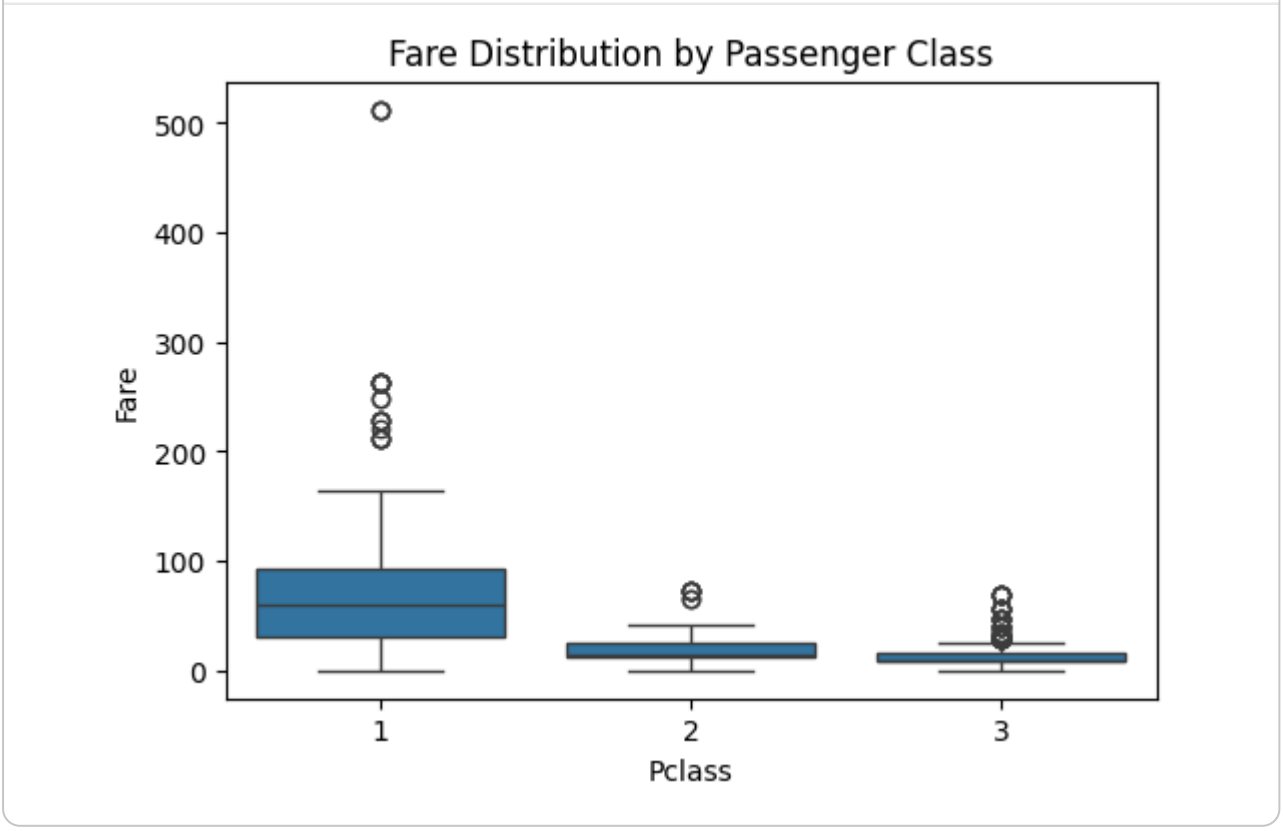
Part G: Scatter Plot – Age vs Fare

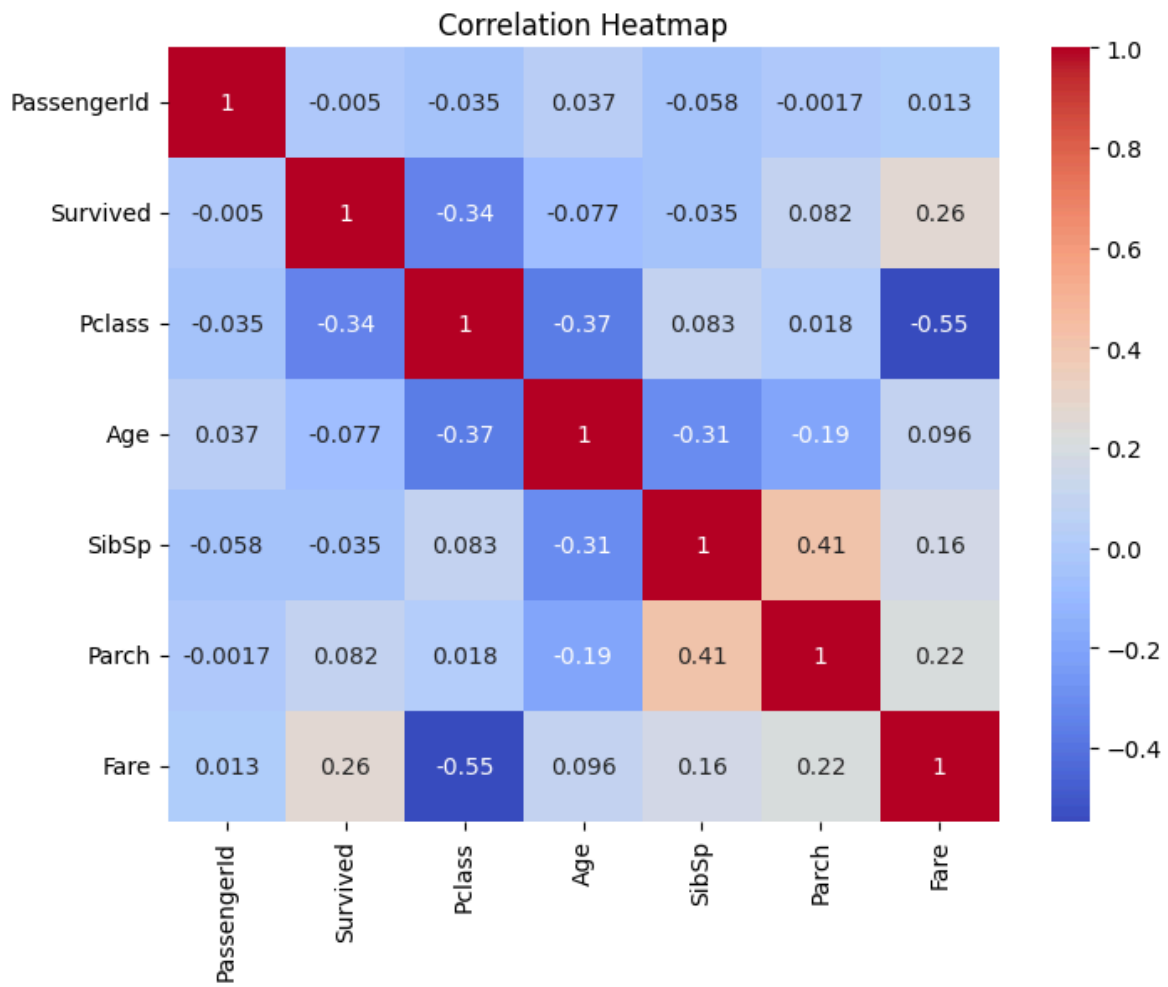
```
plt.figure(figsize=(6,4))
plt.scatter(df["Age"], df["Fare"])
plt.title("Age vs Fare")
plt.xlabel("Age")
plt.ylabel("Fare")
plt.show()
```



Part H: Boxplot – Fare by Passenger Class

```
plt.figure(figsize=(6,4))
sns.boxplot(x="Pclass", y="Fare", data=df)
plt.title("Fare Distribution by Passenger Class")
plt.show()
```



Part I: Correlation Heatmap

```
plt.figure(figsize=(8,6))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
```